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Written Assignment - 2

Q.1 Explain the basic architecture of Generative Adversarial Network.

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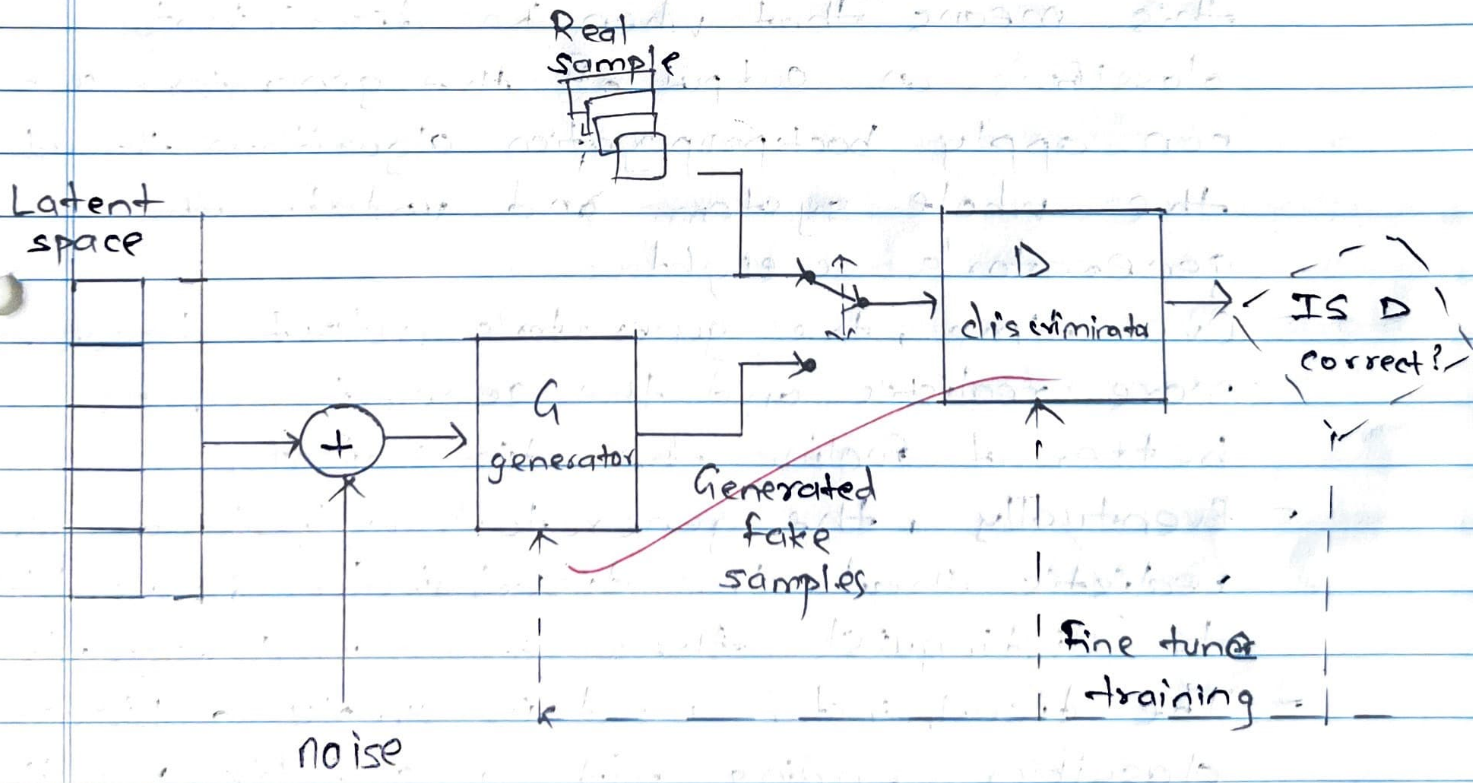
- A generative adversarial network (GAN), is a deep neural network framework which is able to learn from a set of training data and generate new data with the same characteristics as the training data.
- A GAN is made up of two neural network:
  a) The generator, which learns to produce realistic fake data from a random seed.
  b) The discriminator, which learns to distinguish the fake data from realistic data.



- The generator's fake examples, and the training set of real examples, are both fed randomly into the discriminator network.

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- The discriminator does not know whether a particularl input originated from the generator or from the training set.
- To make this Generative and Adversarial process simple, both these blocks are made from DNN based architecture which can be trained through forward and backward propagation techniques.
- Initially, before training has begun, the generator's fake output is very easy for the discriminator to recognize.
- Since the output of the generator is fed directly into the discriminator as input, this means that when the discriminator classifies an output of the generator, we can apply backpropagation algorithm through the whole system and update the generator's weight.
- Over time, the generator's output becomes more realistic and the generator gets better at fooling the discriminator.
- Eventually, the generator's output are so realistic that the discriminator is unable to distinguish them from real example
- The discriminator used is simply a binary classifier, ending with a suitable function such as the softmax function.
- The discriminator outputs an array such as where the two numbers indicate the

discriminator's estimate of the probability of the input example being real or fake.

Q.2 Write a technical article on Deepfake architecture. Include the following points:
i) Introduction
ii) Architecture
iii) Applications.

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1. Introduction

Deepfake technology has gained significant attention due to its ability to manipulate videos, images and audio using deep learning techniques. The term "DeepFake" comes from Deep learning and Fake, referring to synthetic media that can realistically replaces faces or voices in digital content. This technology ~~This dee~~ is built on advanced Generative Adversarial Networks and autoencoders, enabling highly convincing media alterations. While DeepFake has applications in entertainment and creative industries, it also poses ethical and security concern.

2. Deepfake Architecture

Deepfake models typically uses a combination of Autoencoders and GANs to generate synthetic media. The core architecture consists of the following components:

2.1 Autoencoders

Autoencoders are neural networks that learns to encode an input image into a lower dimensional representation and then reconstruct it back. The main components are:

- Encoder: Compresses the input image into a latent space representation.
- Decoder: Reconstructs the image from the latent space representation.

Once trained, the encoder of the source image is combined with decoder of the target image, creating a face swap effect.

2.2 Generative Adversarial Networks

GANs, introduced by IAN Goodfellow in 2014, consists of two neural networks:

- Generator : Creates synthentic images.
- Discriminator : Evaluates whether the image is real or fake.

The generator learns to produce highly realistic Images by competing with the discriminator, which continuously improves in distinguishing real from fake. In Deepfake applications, GANs enhance image realism, smooth transitions, and facial expressions, making the synthetic media indistinguishable from real content.

2.3 Face Alignment and Blending

To ensure realistic results, Deepfake systems incorporate:

- Face Detection & Alignment : Facial landmarks are detected to correctly align faces before training and swapping.
- Seamless Blending : Techniques like Poisson Image Editing and Deep Image Matting help blend the synthetic face naturally with the original background.

2.4 Training Process

Training a DeepFake model involves following steps :

1) Collecting a dataset of source and target faces.
2) Preprocessing images.
3) Training the autoencoders or GAN-based models for several hours or days.
4) Combining the trained models to perform face swapping.
5) Post-processing to enhance realism.

3. Applications

DeepFake technology has various applications, both positive and negative.
Positive applications such as Entertainment & Film industry, Accessibility, Education & research.
Negative applications such as Misinformation and Fake News, Privacy violations, Identity Theft & Fraud.

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